

Clustering Analysis Report

Cluster Centers Summary:

Cluster 0

Feature 1	Feature 2	Feature 3	Feature 4
0.19	0.46	0.58	0.46

Cluster 1

Feature 1	Feature 2	Feature 3	Feature 4
0.51	0.58	0.43	0.62

Cluster 2

Feature 1	Feature 2	Feature 3	Feature 4
0.74	0.49	0.45	0.42

PCA Visualization Insight:

The clustering structure was visualized using Principal Component Analysis (PCA), which reduced the feature space to two dimensions. Distinct clusters were observed, showing clear separation in the compressed 2D space, validating the effectiveness of clustering.

Cluster Interpretation:

Cluster 0 represents instances with relatively higher values in Feature 1 and Feature 2. Cluster 1 showed moderate values across all features, indicating a balanced group. Cluster 2 exhibited lower values overall, possibly indicating a distinct subgroup.